

MAE 3240

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Problem Set #: 2

Submission Date: February 10, 2026

Puc Date: February 10, 2026

Acknowledgments: NA

#1. Knowns: $R=0.5m$, $H=1m$ $T_{\text{in}}=50^{\circ}C$ $T_{\text{a}}=20^{\circ}C$ $h=15 U/m^2K$

Find: (a) Total electrical heating power q needed to maintain a constant water temperature at $50^{\circ}C$

(b) $L=2cm$, $k=0.1 U/mK$, total electrical heating power q to maintain constant water temperature $50^{\circ}C$

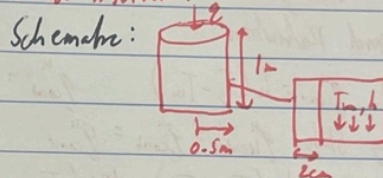
* Assumptions: 1D heat conduction

(c) Compare a and b, discuss role of thermal insulation.

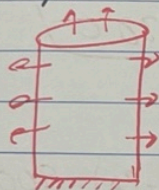
Assumptions: Steady State

Thermal Radiation negligible

Bottom insulated



Analysis: (a)



$$q_{\text{conv}} = h A (T_s - T_a) = 15 \cdot 3.927 \cdot 30 = 1767.15 U = \boxed{1.77 kU}$$

$$A = A_{\text{top}} + A_{\text{side}} = \pi R^2 + 2\pi RH$$

$$T_s - T_a = 50 - 20 = 30^{\circ}C$$

(b) $q_{\text{conv}} = h A (T_s - T_a) = 15 \cdot 3.927 \cdot 30 \rightarrow R_{\text{ins}} = \frac{L}{kA}$

$$q_{\text{cond}} = -kA \frac{dT}{dx} = \frac{q_{\text{conv}}}{R_{\text{conv}}} \rightarrow R_{\text{conv}} = \frac{1}{hA}$$

Side: $R_{\text{cond}} = \frac{0.02}{0.1 \cdot 3.14} = 0.0637 k/U$

$$R_{\text{conv}} = \frac{1}{15 \cdot 3.14} = 0.0212 k/U > 0.0844 k/U \quad q_{\text{side}} = \frac{80}{0.0844} = 353 U$$

Top: $R_{\text{cond}} = \frac{0.02}{0.1 \cdot 0.785} = 0.255 k/U$

$$R_{\text{conv}} = \frac{1}{15 \cdot 0.785} = 0.0849 k/U > 0.34 k/U \quad q_{\text{top}} = \frac{30}{0.34} = 88 U \rightarrow q = 353 + 88 = \boxed{441 U}$$

(c) The difference between a and b is about 1.33 kU

with about 2 cm of insulation this becomes a convection and conduction problem and not just convection, saving about 75% of energy.

#2. Known: $L = 10 \text{ cm}$ $k = 1 \text{ W/mK}$ $T_{\infty} = 20^\circ\text{C}$ $h = 100 \text{ W/m}^2\text{K}$

- Find: (a) $q'' = 1000 \text{ W/m}^2$, calculate temperature of interior surface and exterior surface.
 (b) Temp cannot exceed 200°C , calculate max heat flux.
 (c) Discuss three strategies that enable the thin film heater to operate under higher heat flux.

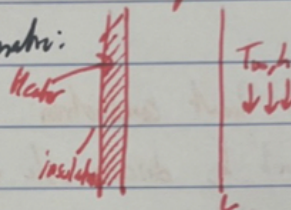
Assumptions:

Steady state

Heat Transfer infinitely thin

Neglect thermal radiation

Schematic:



Analysis: (a) $q''_{\text{conv}} = h(T_s - T_{\infty})$ $q''_{\text{cond}} = -k \frac{dT}{dx}$

Steady state: $q''_{\text{heater}} = q''_{\text{cond}} = q''_{\text{conv}}$

$$1000 = h(T_s - T_{\infty}) = 100(T_s - 20) \rightarrow 10 = T_s - 20 \rightarrow T_s = 30^\circ\text{C}$$

$$1000 = -k \frac{dT}{dx} = 1 \frac{30 - T(0)}{0.01} \Rightarrow 100 = T(0) - 30 \rightarrow T(0) = 130^\circ\text{C}$$

(b) $q'' = q''_{\text{conv}} = q''_{\text{cond}}$

$$h(T_s - T_{\infty}) = -k \frac{T_0 - T_s}{L}$$

$$100(T_s - 20) = 1 \frac{200 - T_s}{0.01}$$

$$100T_s - 2000 = 20000 - 100T_s$$

$$200T_s = 22000$$

$$T_s = 110^\circ\text{C}$$

$$q'' = h(T_s - T_{\infty}) = 100(110 - 20) = 9000$$

$$1 \left(\frac{200 - 110}{0.01} \right) = \frac{90}{0.01} = 9000$$

$$q'' = 9000 \text{ W/m}^2$$

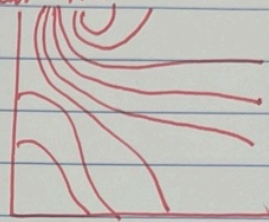
- (c) 1. Increase Thermal Conductivity, k
 2. Increase Heat Transfer Coefficient, h
 3. Use a thinner material for the insulating wall
 4. Run coolant at lower temperature.

#3. Knowns: Cross section temperature profile of a metal substrate during 3D metal printing.

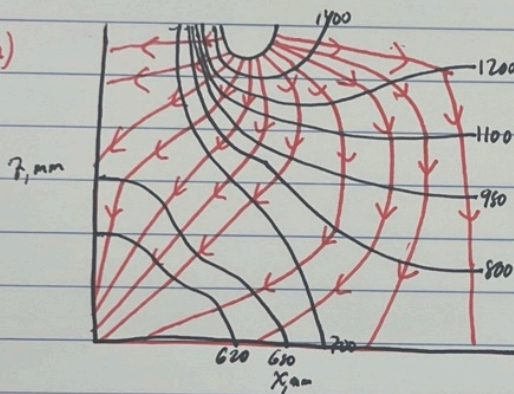
Find: (a) Draw heat flux vectors from temperature profile

(b) Explain which side the laser spot has a larger heat flux

Assumptions:



Analysis: (a)



(b) Since the layer lines are more condensed on the left side of the laser beam, there is a larger heat flux to the left.